

DSA0601 Data Handling and Visualization

Slot A

CO2 – Assessment Test 1 (AT1)

Chart Construction Test

Instructions:

For each question, write the program in the specified language (R or Python), generate the required

visualization, and answer the interpretation sub-questions clearly with justification.

CO2_AT1 – Chart Construction Test

Integrated Disciplines

Each question combines **Data Visualization + R Programming + another discipline** to promote interdisciplinary learning.

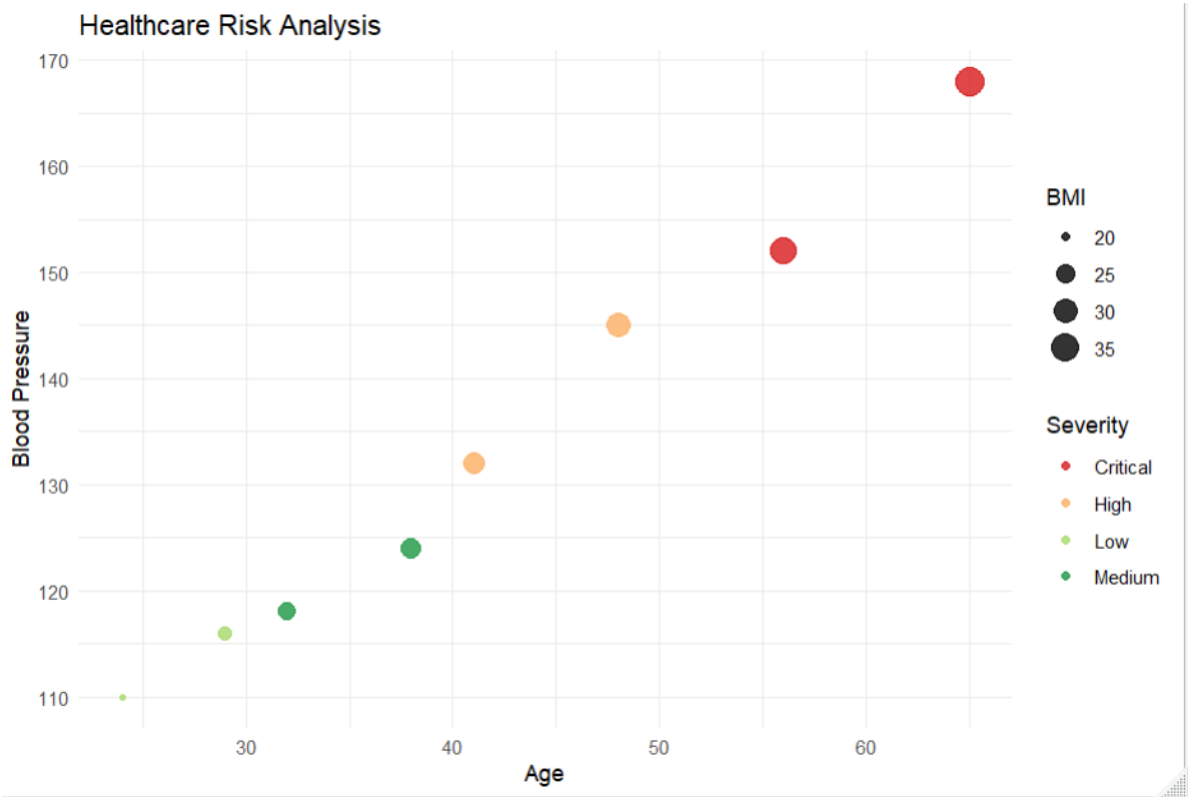
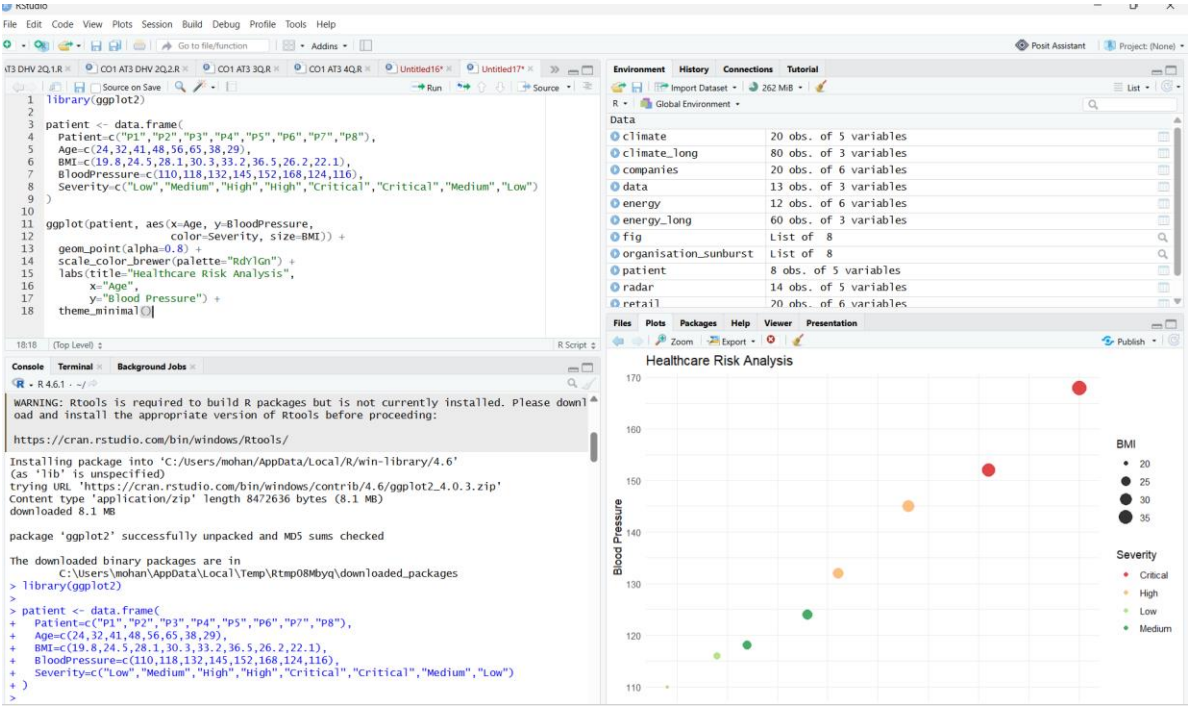
Question 1 (Healthcare Analytics)

Dataset

P1				e BMI Blood Pressure Disease Severity Low Medium High High Critical Critical Medium Low
P2	24	19.8	110	
P3				
P4	32	24.5	118	
P5	41	28.1	132	
P6				
P7	48	30.3	145	
P8	56	33.2	152	
	65	36.5	168	
	38	26.2	124	
	29	22.1	116	

Task

Construct a visualization in R (ggplot2) that:



- Maps **Age** to the x-axis.
- Maps **Blood Pressure** to the y-axis.
- Represents **Disease Severity** using an appropriate color palette.
- Represents **BMI** using point size.
- Justify why your aesthetic mappings and scales effectively communicate health risk.

Answer:

- Age on the x-axis and blood pressure on the y-axis clearly show health risk trends.
- Disease severity uses different colors for easy identification, while BMI point size highlights obesity-related risks.

Interdisciplinary Integration: Healthcare + Medical Statistics + Data Visualization

Question 2 (Climate Science)

Dataset

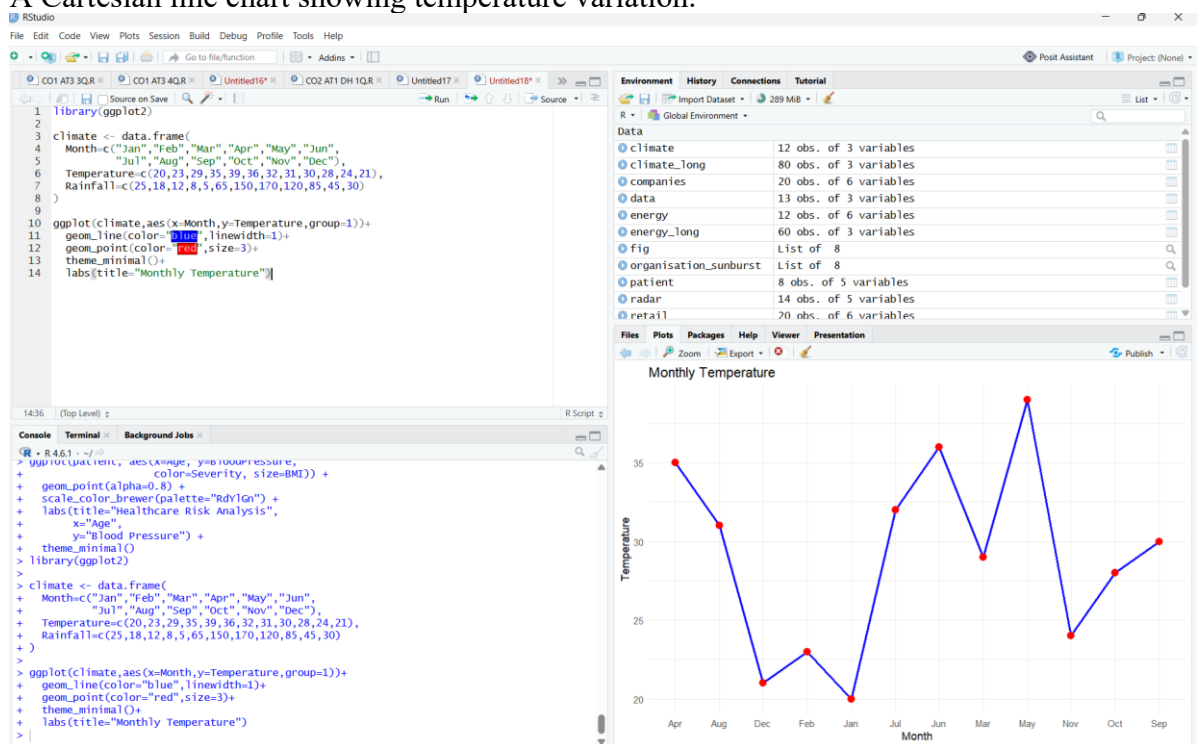
		Radiation (°C)	Rainfall
Jan			25
Feb			18
Mar	180	20	12
Apr	200	23	8
May			5
Jun	240	29	65
Jul	280	35	150
Aug			170
Sep	320	39	120
Oct	300	36	85
Nov	260	32	45

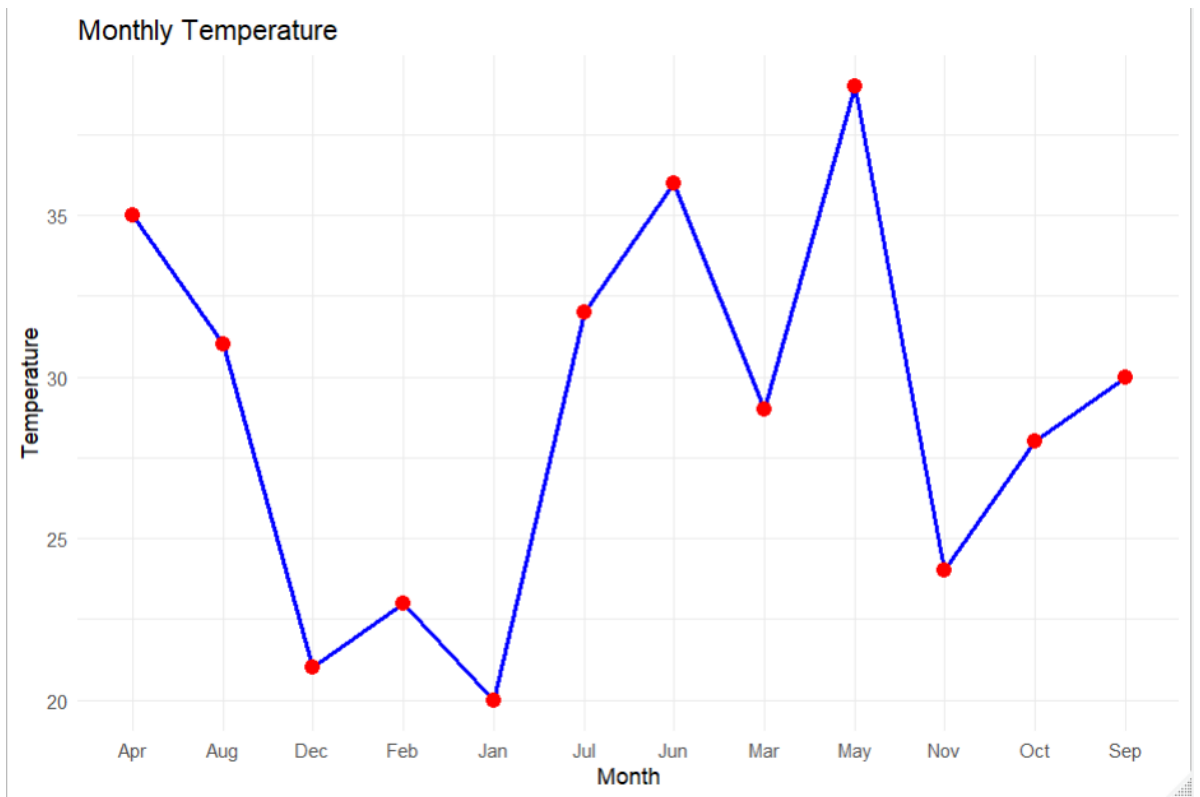
Dec	250	31	30
	240	30	
	220	28	
	190	24	
	170	21	

Task

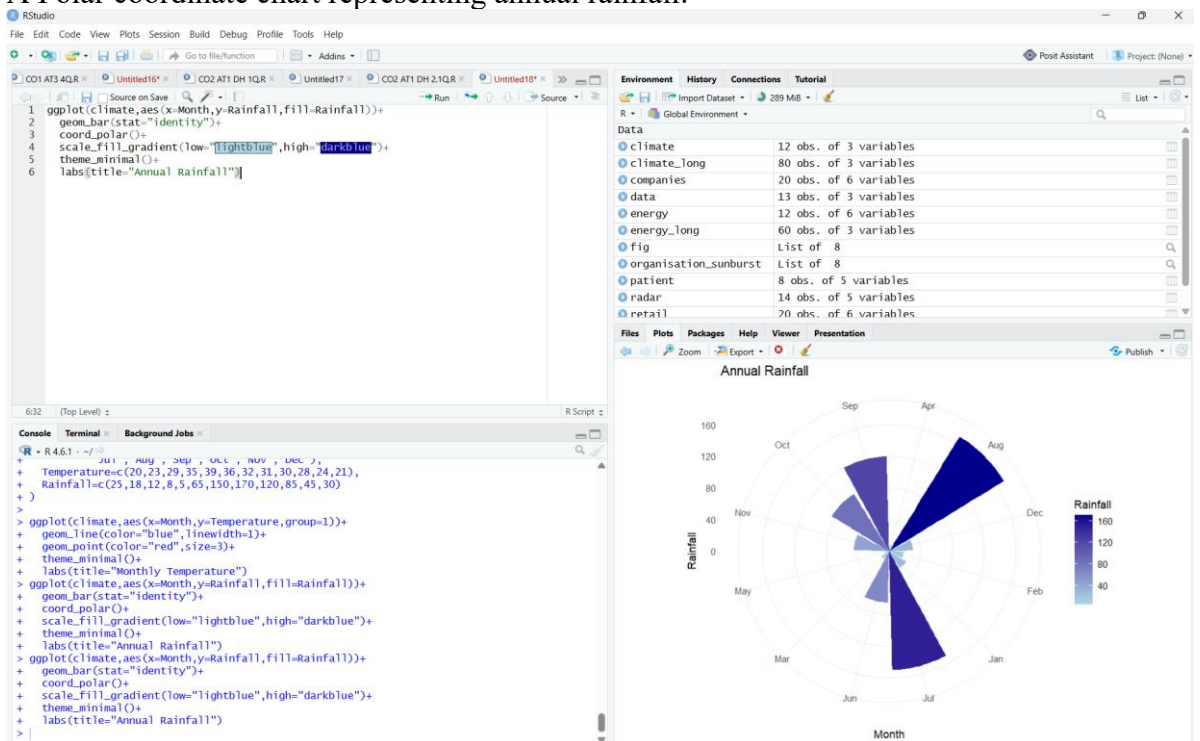
Develop **two visualizations**:

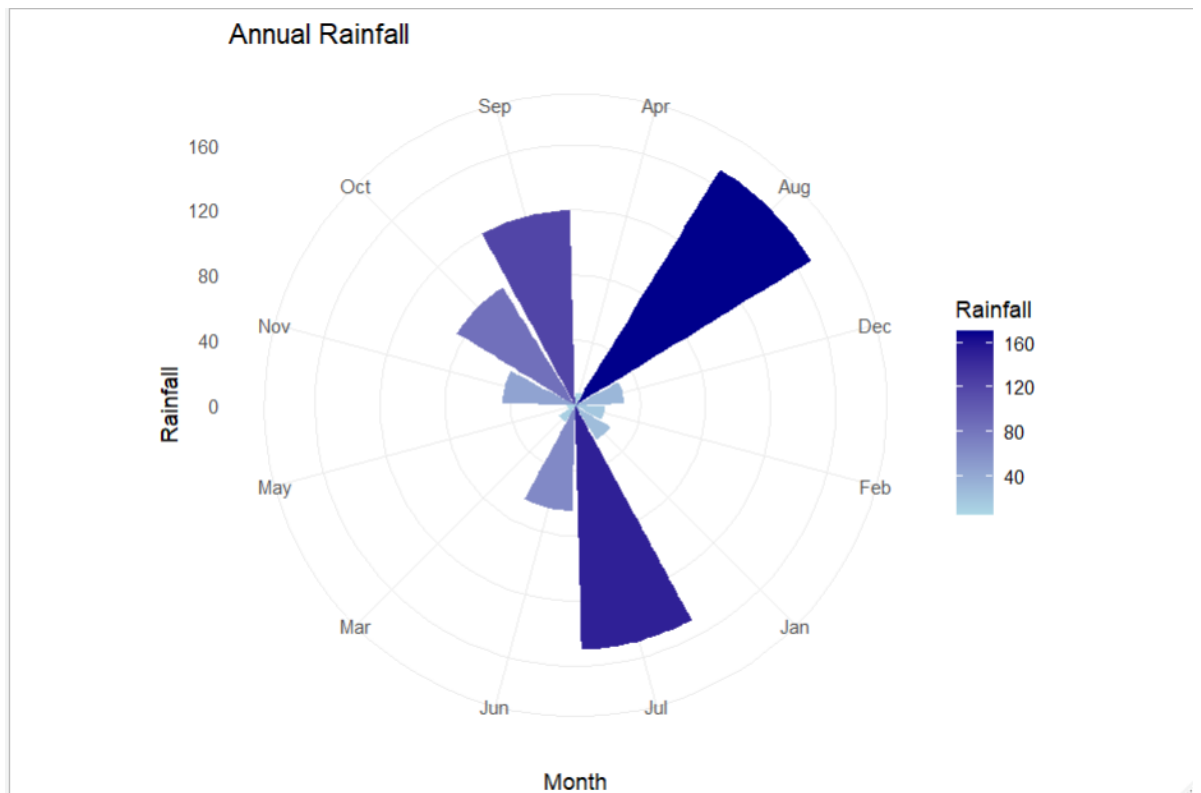
1. A Cartesian line chart showing temperature variation.





2. A Polar coordinate chart representing annual rainfall.





Compare the advantages and limitations of Cartesian and Polar coordinate systems for climate visualization.

Answer:

Cartesian Chart

- Best for showing monthly temperature trends and exact comparisons.
- Easy to identify increases and decreases over time.

Polar Chart

- Best for displaying seasonal rainfall patterns in a circular format.
- Less suitable for comparing exact rainfall values.

Interdisciplinary Integration: Environmental Science + Mathematics + Data Visualization

Question 3 (Business Analytics & Marketing)

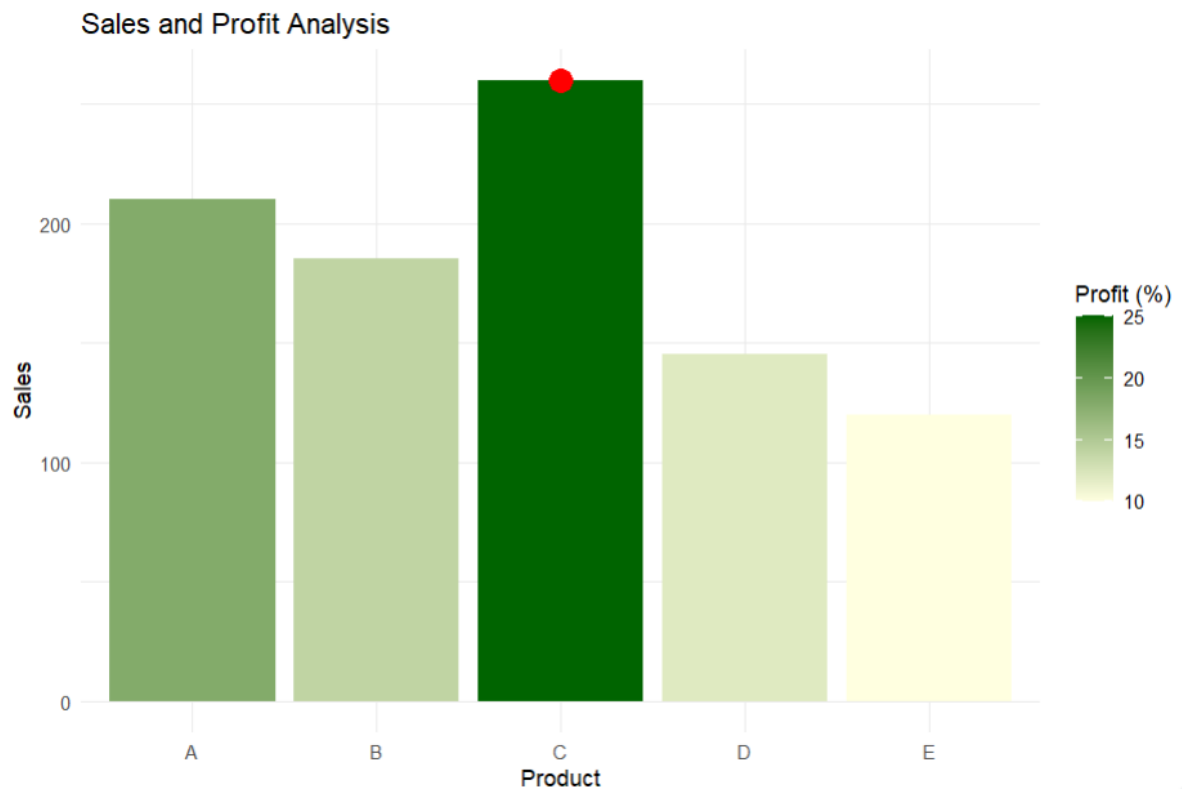
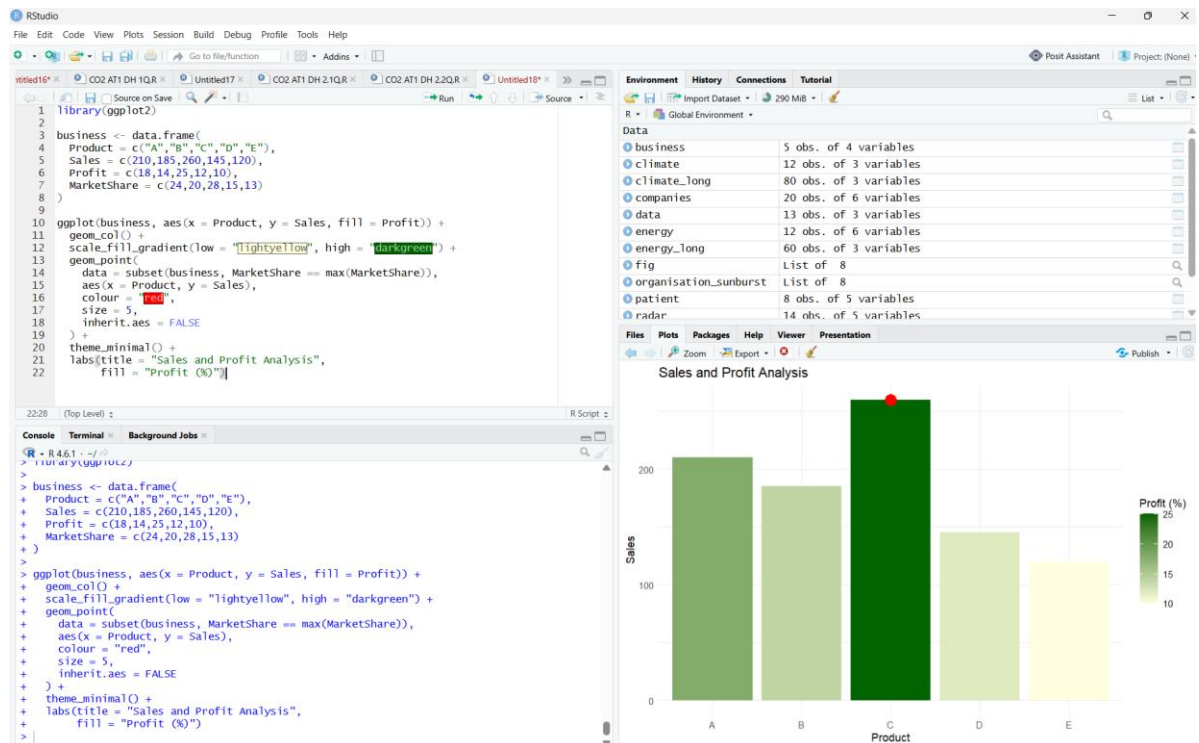
Dataset

A			les (₹ Lakhs) Profit (%) Market
B			Share (%) 24
C	210	18	20
D	185	14	28
E	260	25	15
	145	12	13
	120	10	

Task

Construct a visualization where:

- Sales are represented by bar height.
- Profit percentage is represented using a sequential color scale.
- Highlight the product with the highest market share using a contrasting color.



Explain why your chosen color palette improves business decision-making.

Answer:

- Bar height clearly compares product sales.
- Sequential colors represent increasing profit, while the highest market-share product is highlighted in red for quick business decisions

Interdisciplinary Integration: Marketing + Business Intelligence + Visualization

Question 4 (Transportation Engineering)

Dataset

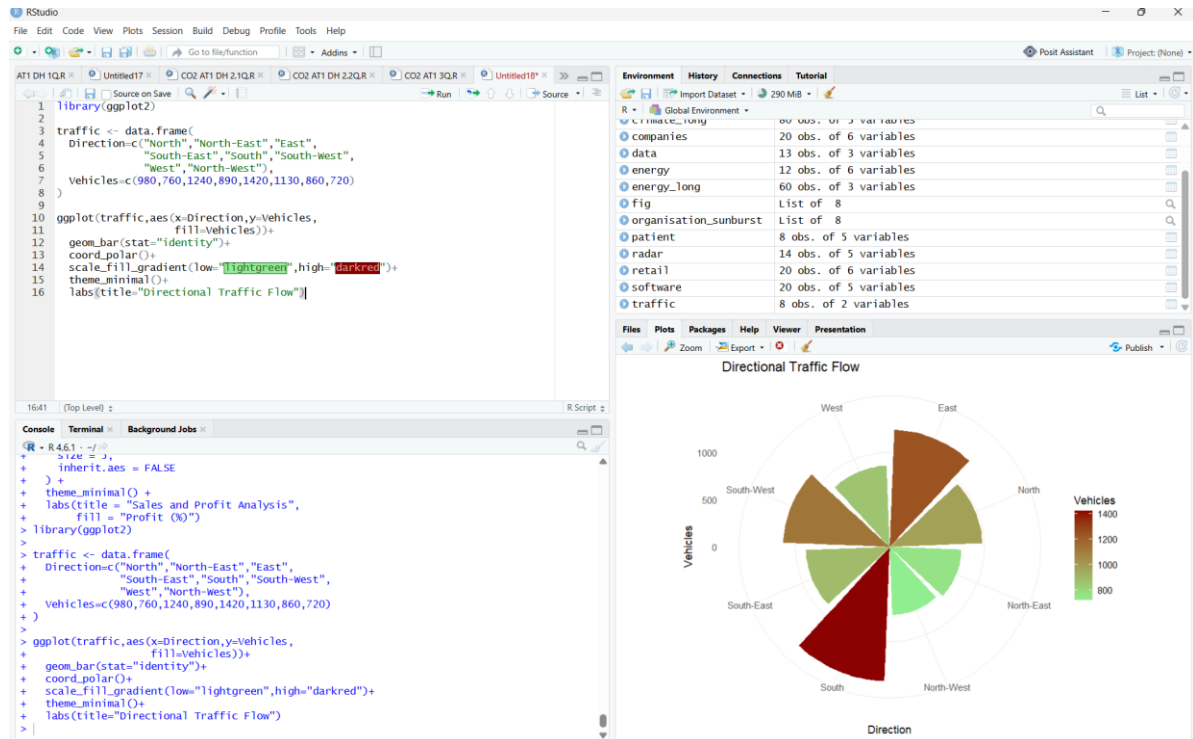
Direction	Vehicles/hour 980
North	1240
North-East 760	
East	1420
South-East 890	860
South	
South-West 1130	
West	

Direction North-West 720	Vehicles/hour
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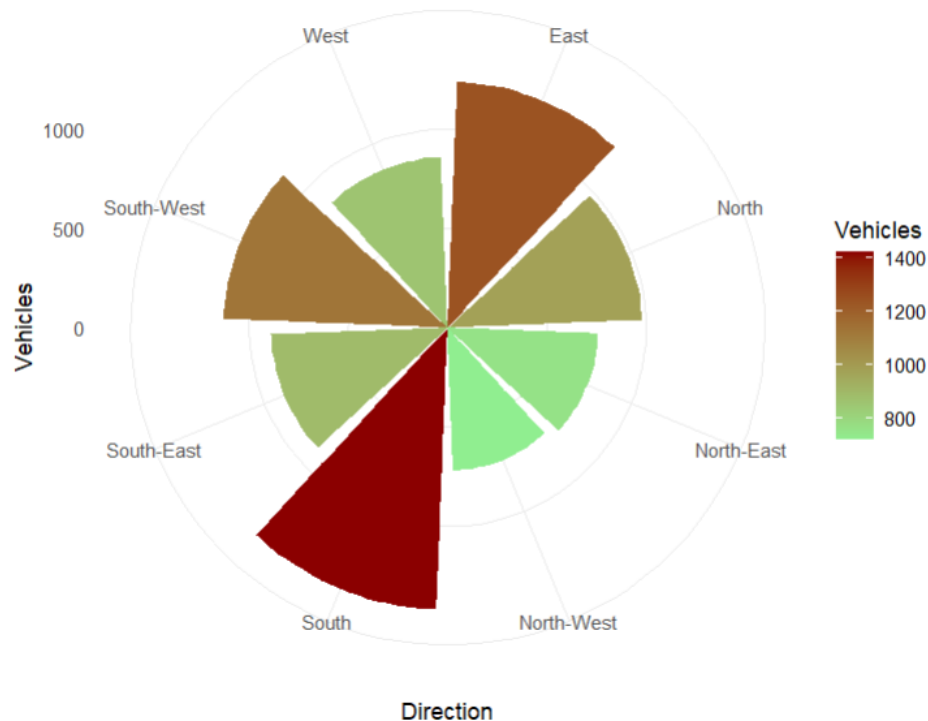
Task

Using **R Programming**:

- Construct a **Polar Coordinate Chart** representing traffic flow.
- Apply an appropriate color palette to distinguish traffic density.



Directional Traffic Flow



- Explain why a Polar chart is more suitable than a Cartesian bar chart for directional traffic analysis.

Answer:

- Polar charts naturally represent compass directions.
- They make directional traffic density easier to visualize than Cartesian bar charts.

Interdisciplinary Integration: Civil Engineering + GIS + Transportation Analytics

Question 5 (Environmental Sustainability Dashboard)

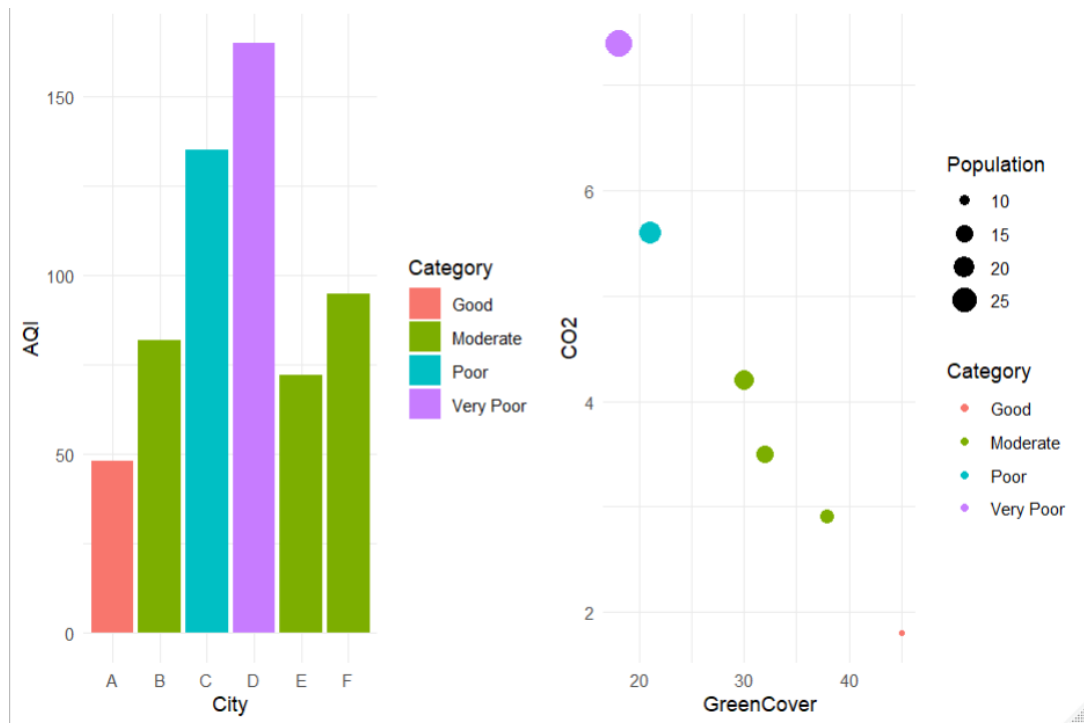
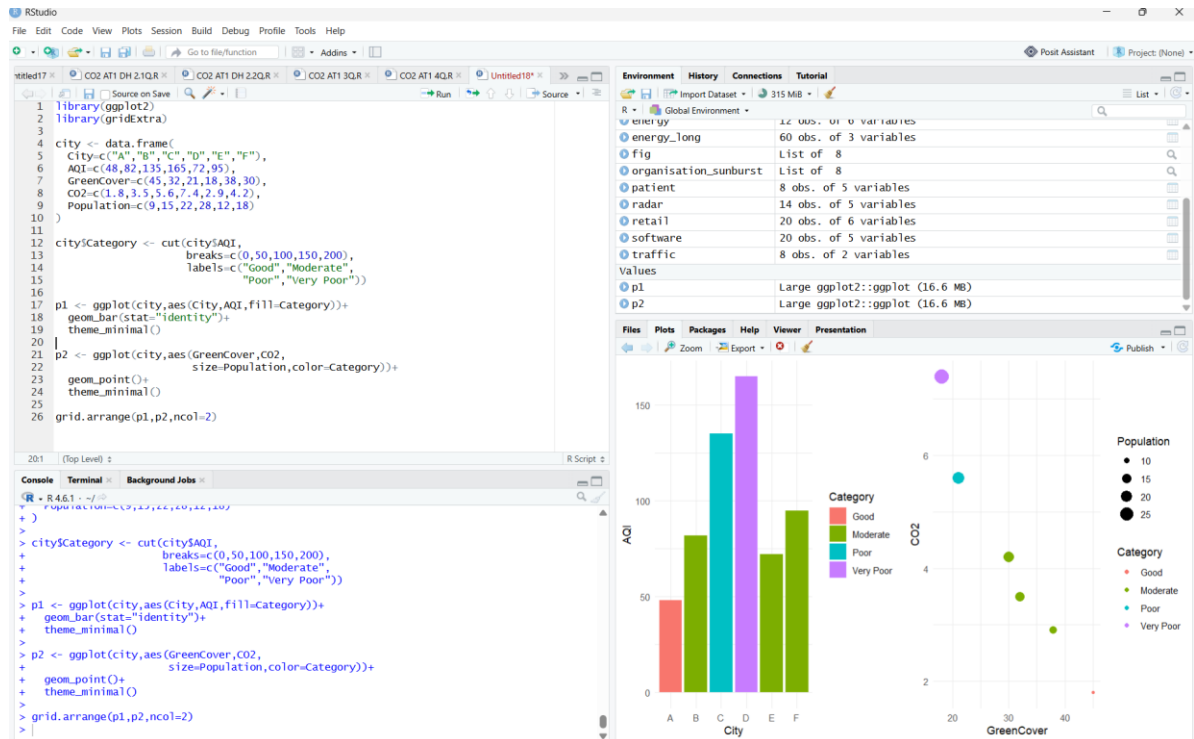
Dataset

A				Green Cover (%) CO₂ Emission Population (Lakhs) 9 15 22 28 12 18
B	48	45	1.8	
C				
D	82	32	3.5	
E	135	21	5.6	
F				
	165	18	7.4	
	72	38	2.9	
	95	30	4.2	

Task

Design a visualization dashboard that includes:

- Appropriate aesthetic mappings.
- Suitable scales for continuous variables.
- An effective color palette for AQI categories.



• Justification for every visualization design choice.

Answer:

- AQI categories use colors to indicate pollution levels clearly.
- Scatter plots show the relationship between green cover, CO₂ emissions, and population, making the dashboard easy to interpret.